Results

3.1 Socio-demographic Characteristics

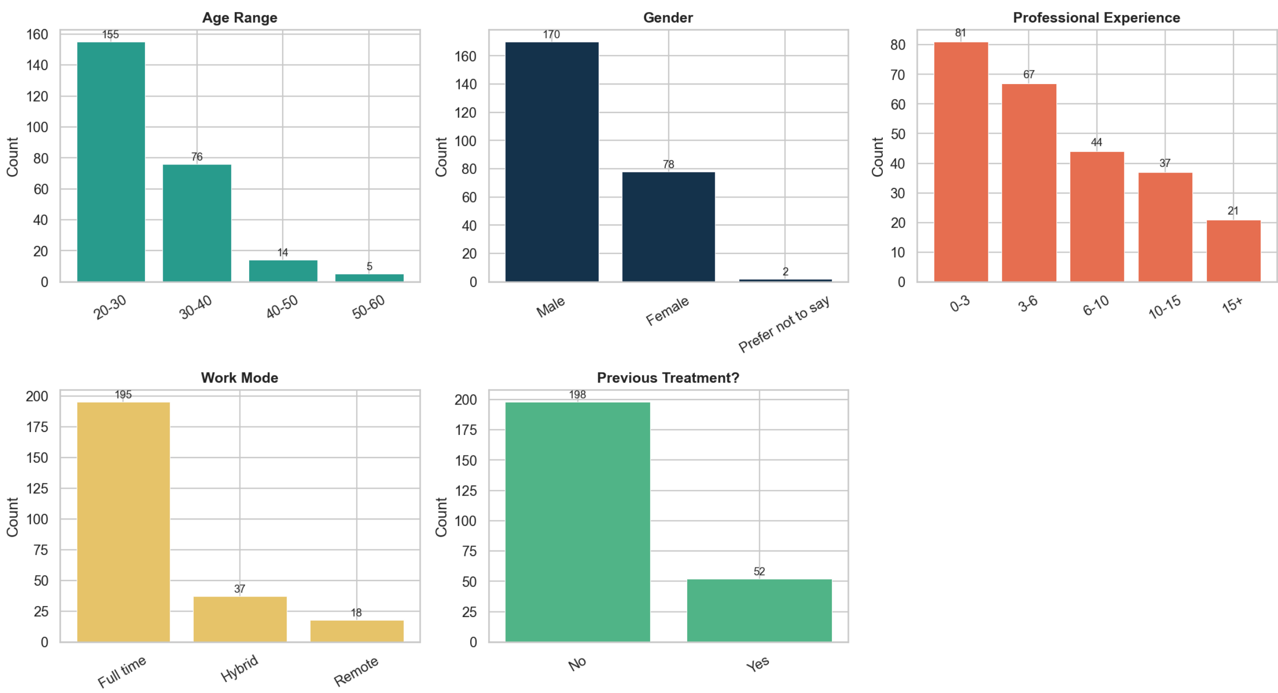
A total of 250 IT professionals from Bengaluru participated in this study. The majority of respondents were in the 20-30 years age group (62.0%), and 68.0% were Male. Regarding professional experience, the distribution across categories is shown in Table 1. 78.0% worked full-time. Only 20.8% had received prior periodontal treatment.

Table 1. Socio-demographic characteristics of respondents (N=250).

Variable	Category	n	%
Age Range	20-30 years	155	62.0
	30-40 years	76	30.4
	40-50 years	14	5.6
	50-60 years	5	2.0
Gender	Male	170	68.0
	Female	78	31.2
	Prefer not to say	2	0.8
	Other	0	0.0
Professional Experience	0-3 years	81	32.4
	3-6 years	67	26.8
	6-10 years	44	17.6
	10-15 years	37	14.8
	15+ years	21	8.4
Work Mode	Full time	195	78.0
	Hybrid	37	14.8

Variable	Category	n	%
	Work from home (Remote)	18	7.2
Previous Treatment	No	198	79.2
	Yes	52	20.8

Figure 1. Demographic Profile of Respondents (N=250)



3.2 Knowledge Regarding Periodontal Health

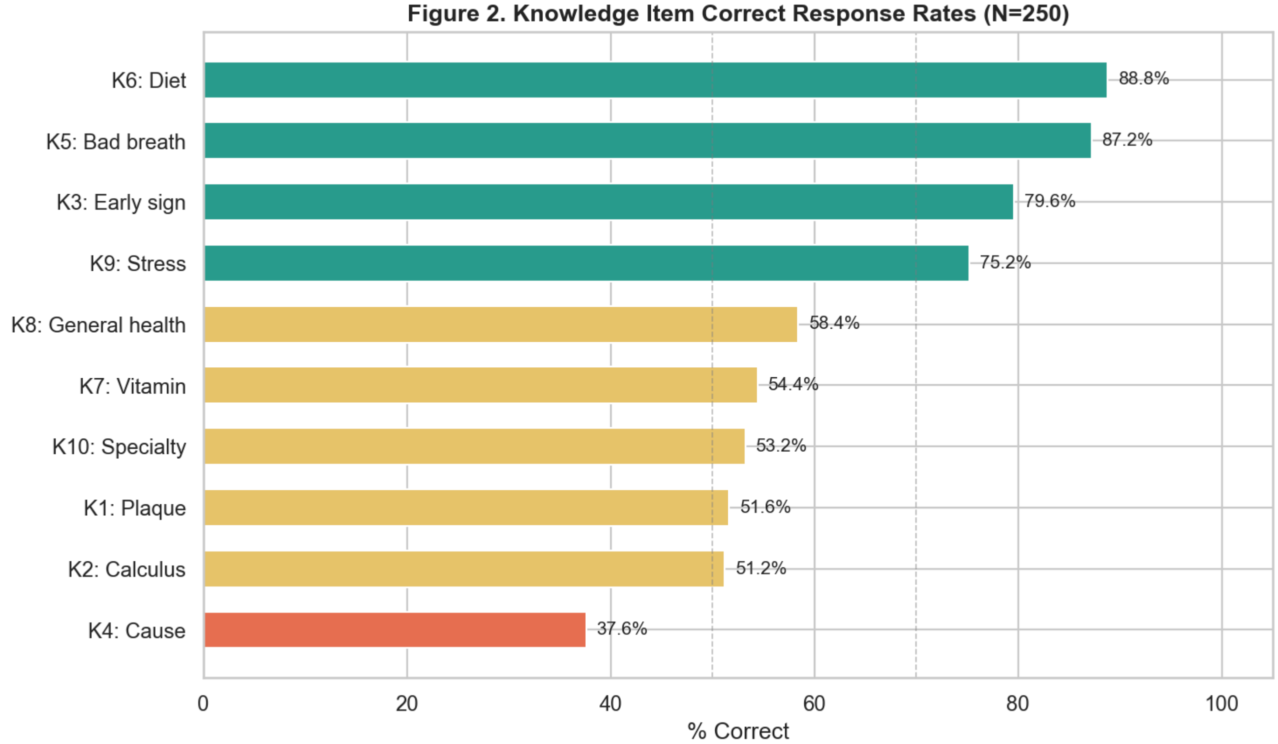
The mean knowledge score was 6.37 ± 2.36 out of 10. The highest correct response rate was observed for K6 (Diet) at 88.8%, while the lowest was for K4 (Cause) at 37.6%. Table 2 provides the item-level breakdown.

Table 2. Knowledge item analysis (N=250).

Item	Topic	Correct Answer	n Correct	% Correct
K1	Plaque	A soft film of bacteria and food debris on the teeth	129	51.6
K2	Calculus	Hard mineral deposit on the teeth	128	51.2

Item	Topic	Correct Answer	n Correct	% Correct
K3	Early sign	Bleeding gums while brushing	199	79.6
K4	Cause	Dental plaque accumulation	94	37.6
K5	Bad breath	Gum disease	218	87.2
K6	Diet	Processed sugary foods and drinks	222	88.8
K7	Vitamin	Vitamin C	136	54.4
K8	General health	It can contribute to heart disease and diabetes	146	58.4
K9	Stress	Development of gum disease due to inflammation	188	75.2
K10	Specialty	Periodontics	133	53.2

Figure 2. Knowledge Item Correct Response Rates (N=250)



3.3 Attitude Towards Periodontal Health

The mean attitude score was 27.26 ± 4.65 out of 40. Items A5 and A8 were reverse-scored as they represent negative attitudes. Table 3 shows the response distribution for each attitude statement.

Table 3. Attitude item analysis (N=250). *Reverse-scored items.

Item	Statement	SD	D	A	SA	Mean ± SD
A1	Neglecting oral hygiene develops gum disease.	41 (16.4%)	82 (32.8%)	83 (33.2%)	44 (17.6%)	2.52 ± 0.97
A2	Bleeding gums should not be ignored.	25 (10.0%)	64 (25.6%)	117 (46.8%)	44 (17.6%)	2.72 ± 0.87
A3	Attending workplace awareness programs will improve gum health.	14 (5.6%)	81 (32.4%)	121 (48.4%)	34 (13.6%)	2.70 ± 0.77
A4	Maintaining oral hygiene is equally important as maintaining physical fitness.	32 (12.8%)	49 (19.6%)	120 (48.0%)	49 (19.6%)	2.74 ± 0.92
A5	Visiting a dentist is necessary only when I have pain or discomfort.*	37 (14.8%)	132 (52.8%)	59 (23.6%)	22 (8.8%)	2.74 ± 0.82
A6	Poor gum health can negatively affect my confidence or social interactions.	14 (5.6%)	65 (26.0%)	123 (49.2%)	48 (19.2%)	2.82 ± 0.80
A7	Oral hygiene aids like floss or mouthwash can be used if recommended by a dentist.	17 (6.8%)	66 (26.4%)	136 (54.4%)	31 (12.4%)	2.72 ± 0.77
A8	Oral hygiene maintenance is time consuming and difficult to follow.*	35 (14.0%)	126 (50.4%)	68 (27.2%)	21 (8.4%)	2.70 ± 0.81

Item	Statement	SD	D	A	SA	Mean ± SD
A9	Workplace stress can have an impact on my gum and oral health.	17 (6.8%)	68 (27.2%)	134 (53.6%)	31 (12.4%)	2.72 ± 0.77
A10	Investing in preventive dental care is better than spending on treatment later.	20 (8.0%)	48 (19.2%)	123 (49.2%)	59 (23.6%)	2.88 ± 0.86

** Reverse-scored items (A5, A8): scores are inverted so that higher values indicate a more positive attitude.*

3.4 Practice of Oral Hygiene

The mean practice score was 20.96 ± 4.27 out of 30. Table 4 details the response distribution and mean score for each practice item.

Table 4. Practice item analysis (N=250).

Item	Topic	Option A	Option B	Option C	Option D	Mean Score ± SD
P1	Brushing frequency	141 (56.4%)	7 (2.8%)	98 (39.2%)	4 (1.6%)	2.14 ± 1.00
P2	Brushing direction	91 (36.4%)	34 (13.6%)	34 (13.6%)	91 (36.4%)	1.73 ± 1.10
P3	Brush softness	138 (55.2%)	44 (17.6%)	64 (25.6%)	4 (1.6%)	2.26 ± 0.90
P4	Brushing timing	111 (44.4%)	5 (2.0%)	127 (50.8%)	7 (2.8%)	2.04 ± 0.99
P5	Brushing duration	20 (8.0%)	181 (72.4%)	35 (14.0%)	14 (5.6%)	2.51 ± 0.92
P6	Post-meal rinse	151 (60.4%)	75 (30.0%)	13 (5.2%)	11 (4.4%)	2.46 ± 0.81

Item	Topic	Option A	Option B	Option C	Option D	Mean Score ± SD
P7	Cleaning aids	104 (41.6%)	43 (17.2%)	73 (29.2%)	30 (12.0%)	1.12 ± 0.97
P8	Brush replacement	159 (63.6%)	53 (21.2%)	6 (2.4%)	32 (12.8%)	2.38 ± 0.90
P9	Dental visit	40 (16.0%)	98 (39.2%)	23 (9.2%)	89 (35.6%)	1.72 ± 1.15
P10	Risk habits	15 (6.0%)	203 (81.2%)	22 (8.8%)	10 (4.0%)	2.60 ± 0.88

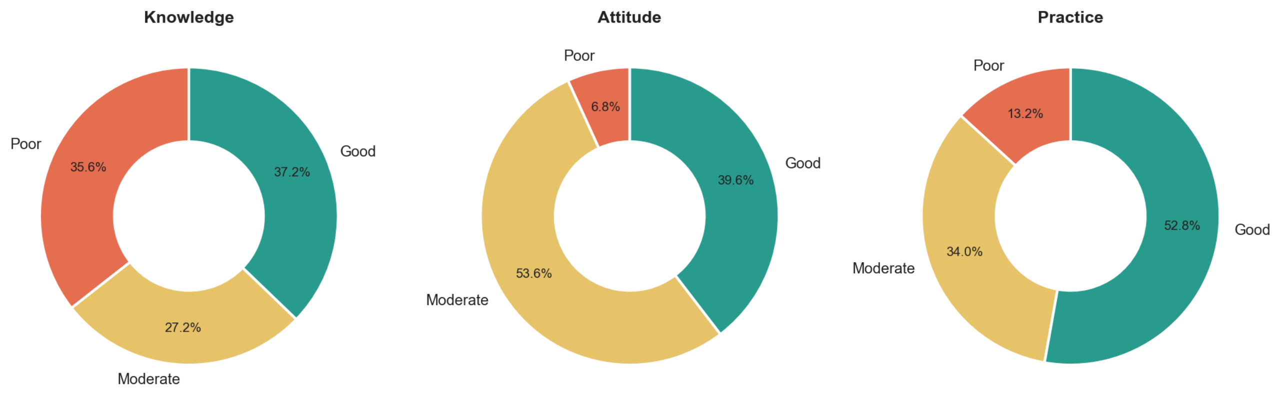
3.5 Distribution of KAP Levels

Using Bloom's cut-off points (Poor <50%, Moderate 50–69%, Good ≥70%), the distribution of KAP levels was as follows: knowledge (Good: 37.2%, Moderate: 27.2%, Poor: 35.6%); attitude (Good: 39.6%, Moderate: 53.6%, Poor: 6.8%); practice (Good: 52.8%, Moderate: 34.0%, Poor: 13.2%). See Table 5 and Figure 3.

Table 5. Distribution of KAP levels (N=250).

Domain	Poor (<50%)	Moderate (50–69%)	Good (≥70%)
Knowledge	89 (35.6%)	68 (27.2%)	93 (37.2%)
Attitude	17 (6.8%)	134 (53.6%)	99 (39.6%)
Practice	33 (13.2%)	85 (34.0%)	132 (52.8%)

Figure 3. Distribution of KAP Levels (N=250)



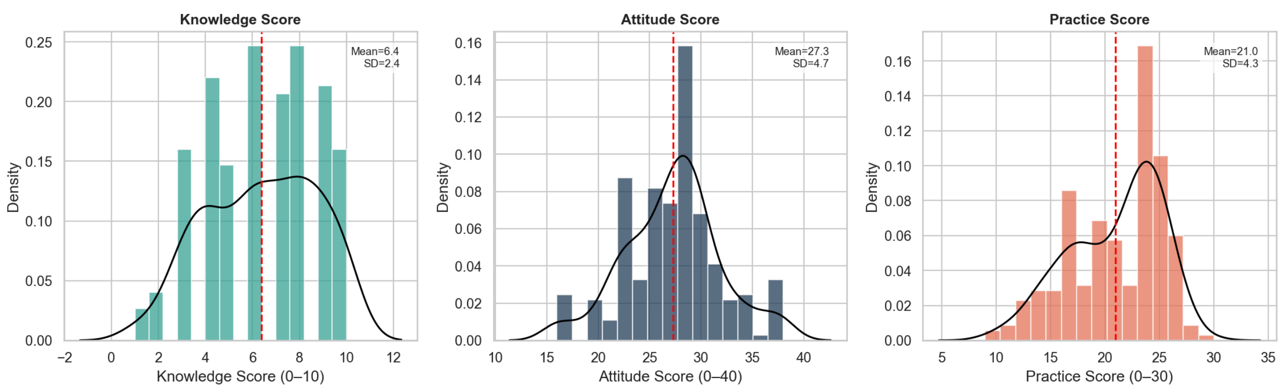
3.6 Descriptive Statistics of KAP Scores

Table 6 summarizes the descriptive statistics for each KAP domain. The median knowledge score was 6, attitude score was 28, and practice score was 22.

Table 6. Descriptive statistics of KAP scores (N=250).

Domain	Possible Range	Mean	SD	Median	Min	Max
Knowledge	0–10	6.37	2.36	6.0	1	10
Attitude	0–40	27.26	4.65	28.0	16	38
Practice	0–30	20.96	4.27	22.0	9	30

Figure 4. Distribution of KAP Scores (N=250)



3.7 Correlation Between Knowledge, Attitude, and Practice

Spearman's correlation analysis revealed a weak positive correlation between knowledge and attitude (ρ=0.316, p=<0.001), a weak positive correlation between knowledge and practice (ρ=0.379, p=<0.001), and a weak positive correlation between attitude and practice (ρ=0.240, p=<0.001). This suggests that higher knowledge is associated with more positive attitudes and better practices.

Table 7. Inter-KAP Spearman correlations (N=250).

Pair	Spearman ρ	p-value	Sig.
Knowledge vs Attitude	0.316	0.0000	***
Knowledge vs Practice	0.379	0.0000	***
Attitude vs Practice	0.240	0.0001	***

Figure 5. Inter-KAP Correlations (N=250)

3.8 Association of Demographic Variables with KAP Scores

Statistically significant differences (p<0.05) were observed for: knowledge scores by Gender (p=0.0000); practice scores by Professional Experience (p=0.0012); attitude scores by Work Mode (p=0.0029). No significant differences were found for other demographic comparisons. Table 8 provides the full results.

Table 8. Comparison of KAP scores by demographic variables (N=250).

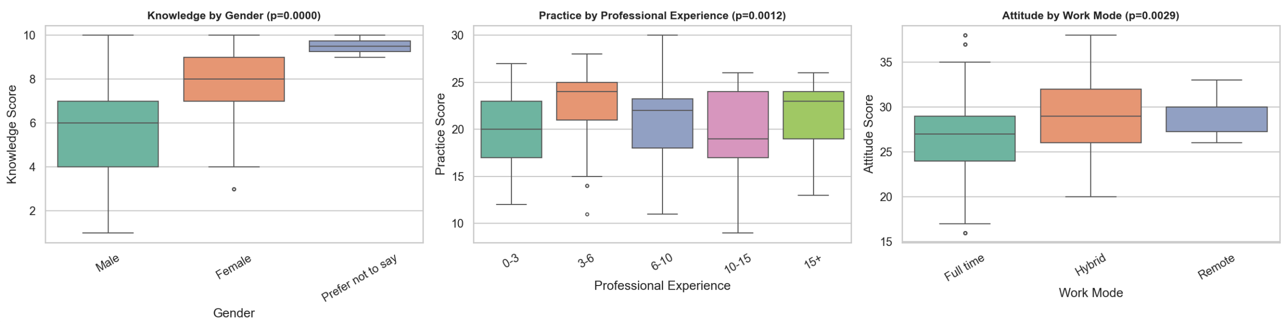
Variable	Score	Group Mean±SD	Test	Statistic	p-value	Sig.

Variable	Score	Group Mean±SD	Test	Statistic	p-value	Sig.
Age Range	Knowledge	20-30 years: 6.3±2.5; 30-40 years: 6.5±2.0; 40-50 years: 6.1±2.7; 50-60 years: 6.2±0.4	Kruskal-Wallis H	0.5	0.9153	
	Attitude	20-30 years: 28.0±3.9; 30-40 years: 25.9±5.7; 40-50 years: 26.6±5.0; 50-60 years: 26.8±3.6	Kruskal-Wallis H	7.5	0.0568	
	Practice	20-30 years: 21.3±4.3; 30-40 years: 20.4±4.3; 40-50 years: 20.7±3.4; 50-60 years: 18.2±4.4	Kruskal-Wallis H	4.6	0.2019	
Gender	Knowledge	Male: 5.7±2.2; Female: 7.7±2.0; Prefer not to say: 9.5±0.7	Kruskal-Wallis H	42.1	0.0000	***
	Attitude	Male: 26.7±4.5; Female: 28.4±4.8; Prefer not to say: 30.0±0.0	Kruskal-Wallis H	5.7	0.0570	
	Practice	Male: 20.7±4.3; Female: 21.5±4.2; Prefer not to say: 22.0±0.0	Kruskal-Wallis H	1.8	0.4100	
Professional Experience	Knowledge	0-3 years: 6.4±2.3; 3-6 years: 6.5±2.7; 6-10 years: 6.1±2.1;	Kruskal-Wallis H	8.6	0.0712	

Variable	Score	Group Mean±SD	Test	Statistic	p-value	Sig.
		10-15 years: 7.0±2.1; 15+ years: 5.2±2.2				
	Attitude	0-3 years: 27.7±4.3; 3-6 years: 27.9±3.8; 6-10 years: 26.5±5.4; 10-15 years: 26.6±5.8; 15+ years: 26.2±4.5	Kruskal-Wallis H	3.5	0.4837	
	Practice	0-3 years: 20.1±4.1; 3-6 years: 22.7±3.9; 6-10 years: 20.5±4.7; 10-15 years: 20.0±4.4; 15+ years: 21.0±4.0	Kruskal-Wallis H	18.0	0.0012	**
Work Mode	Knowledge	Full time: 6.3±2.4; Hybrid: 7.1±2.2; Work from home (Remote): 5.8±1.8	Kruskal-Wallis H	4.9	0.0847	
	Attitude	Full time: 26.8±4.7; Hybrid: 28.6±5.0; Work from home (Remote): 29.4±2.1	Kruskal-Wallis H	11.7	0.0029	**
	Practice	Full time: 21.0±4.5; Hybrid: 21.1±3.1; Work from home (Remote): 19.9±3.7	Kruskal-Wallis H	2.1	0.3545	

Variable	Score	Group Mean±SD	Test	Statistic	p-value	Sig.
Previous Treatment	Knowledge	No: 6.5±2.4; Yes: 6.1±2.1	Mann-Whitney U	5741.5	0.1979	
	Attitude	No: 27.3±4.3; Yes: 27.3±5.7	Mann-Whitney U	4505.5	0.1651	
	Practice	No: 20.7±4.3; Yes: 21.8±3.9	Mann-Whitney U	4408.5	0.1096	

Figure 6. KAP Scores by Demographic Variables (p < 0.05)



Discussion

This cross-sectional study assessed the knowledge, attitude, and practice (KAP) regarding periodontal health among 250 IT professionals working in Bengaluru, India. The findings of this study provide valuable insights into the oral health awareness and behaviour of a population that is often overlooked in periodontal research, as most existing KAP studies in dentistry have focused on healthcare workers, university students in health-related disciplines, or the general public rather than information technology workers specifically.

4.1 Knowledge Regarding Periodontal Health

The mean knowledge score in this study was 6.37 ± 2.36 out of 10, with 37.2% of respondents demonstrating good knowledge (≥70%), 27.2% moderate knowledge, and

35.6% poor knowledge. This finding indicates that a substantial proportion of IT professionals have knowledge gaps regarding periodontal health fundamentals. The nearly equal three-way split between good, moderate, and poor knowledge levels is concerning, as it suggests that over one-third of working professionals lack even basic understanding of gum disease and its implications.

At the item level, respondents demonstrated the strongest knowledge regarding the relationship between dietary choices and gum health (K6: Diet, 88.8% correct) and the association between bad breath and gum disease (K5: 87.2% correct). These relatively high correct response rates may be attributed to widespread public health messaging about sugar's detrimental effects on oral health and the intuitive connection between oral conditions and halitosis. In contrast, the poorest knowledge was observed for the primary cause of early gum disease (K4: Cause, 37.6% correct), where most respondents failed to identify dental plaque accumulation as the key aetiological factor. Similarly, the concepts of dental plaque (K1: 51.6%) and calculus (K2: 51.2%) were poorly understood, with approximately half of respondents unable to correctly define these fundamental terms. These findings are consistent with previous studies that have reported deficient knowledge regarding specific oral disease mechanisms among non-healthcare populations [1,2]. The poor understanding of plaque and calculus as distinct entities is particularly noteworthy, as this conceptual confusion may hinder individuals from appreciating the importance of professional dental cleaning and regular scaling.

Knowledge regarding Vitamin C's role in gum health (K7: 54.4%) and the dental specialty of periodontics (K10: 53.2%) was also limited, hovering just above chance levels. The unfamiliarity with periodontics as a dental specialty suggests that many IT professionals may not know where to seek specialised care even if they recognise the need for it. This echoes findings from an earlier study among university students, which highlighted that awareness of dental specialties was significantly lower compared to medical specialties [3]. However, the relatively strong performance on the stress-gum disease connection (K9: 75.2%) is encouraging, particularly given the high-stress nature of IT work environments, as it indicates that many professionals may already recognise workplace stress as a risk factor for periodontal problems.

4.2 Attitude Towards Periodontal Health

The mean attitude score was 27.26 ± 4.65 out of 40, with the majority of respondents (53.6%) demonstrating moderate attitudes and 39.6% showing good attitudes towards periodontal health. Only a small minority (6.8%) had poor attitudes. This finding suggests that while most IT professionals generally hold favourable views about oral health, the predominance of moderate rather than good attitudes indicates room for improvement in translating awareness into stronger convictions.

The attitude item with the highest mean score was A10 (2.88 ± 0.86), reflecting broad agreement that investing in preventive dental care is preferable to spending on treatment later. This is a positive finding, as it suggests that the cost-benefit reasoning common in the IT industry may naturally extend to health prevention decisions. Item A6, regarding the impact of poor gum health on confidence and social interactions (mean 2.82 ± 0.80), also showed relatively high agreement, indicating that IT professionals recognise the social and professional consequences of poor oral health.

Notably, when examining the reverse-scored items, A5 revealed that a combined 32.4% of respondents (23.6% Agree + 8.8% Strongly Agree) believed that visiting a dentist is necessary only when experiencing pain or discomfort. This reactive rather than preventive approach to dental care is a common finding in KAP studies across various populations [4,5] and represents a significant barrier to early detection and treatment of periodontal disease. Similarly, A8 showed that 35.6% of respondents perceived oral hygiene maintenance as time-consuming and difficult to follow, which may reflect the demanding work schedules typical of the IT industry and their potential impact on health-related self-care behaviours.

The finding that respondents generally agreed that workplace awareness programs could improve gum health (A3: 62.0% Agree/Strongly Agree) is practically significant, as it suggests receptivity to employer-sponsored oral health initiatives. This aligns with the growing body of evidence supporting workplace health promotion programs as effective vehicles for improving health literacy among working populations [6].

4.3 Oral Hygiene Practices

The mean practice score of 20.96 ± 4.27 out of 30 was the strongest of the three KAP domains, with 52.8% of respondents classified as having good practices and 34.0% moderate practices. This finding indicates that IT professionals in Bengaluru generally maintain reasonable oral hygiene habits, which is encouraging given the sedentary and often high-stress nature of their work.

Analysis of individual practice items revealed several notable patterns. The majority of respondents (56.4%) reported brushing twice daily (P1), which aligns with the recommended brushing frequency. However, 44.4% brushed only in the morning (P4), missing the critically important night-time brushing that removes the day's accumulated plaque. The most encouraging finding was that 81.2% of respondents did not consume gutka, pan, alcohol, or cigarettes (P10), reflecting relatively low rates of harmful oral habits in this professional population. Additionally, 72.4% brushed for the recommended duration of 1–3 minutes (P5), and 60.4% rinsed their mouth after every meal (P6).

Conversely, several practice areas showed significant deficiencies. The use of interdental cleaning aids (P7) was notably weak, with a mean score of only 1.12 ± 0.97 — the lowest among all practice items. While 41.6% used tongue scrapers, only 17.2% used dental floss, and 29.2% relied on toothpicks, which is not considered an optimal cleaning aid. This finding is consistent with previous studies reporting low rates of flossing and interdental cleaning even among populations with otherwise adequate brushing habits [7,8]. Dental visit patterns (P9) were also concerning: 16.0% had never visited a dentist, and 35.6% had not visited in over a year, resulting in a low mean score of 1.72 ± 1.15 for this item. The infrequent dental visits, despite 79.2% never having received periodontal treatment, suggest that many professionals may only seek dental care reactively, corroborating the attitudinal finding regarding pain-driven dental visits.

4.4 Correlation Between Knowledge, Attitude, and Practice

The Spearman's correlation analysis demonstrated statistically significant positive correlations between all three KAP domains ($p<0.001$). The correlation between knowledge and practice ($\rho=0.379$) was the strongest, followed by knowledge and attitude ($\rho=0.316$), and attitude and practice ($\rho=0.240$). Although the correlation coefficients indicate weak positive associations, they are all highly significant and suggest meaningful relationships between these domains.

The finding that knowledge-practice correlation was stronger than knowledge-attitude or attitude-practice correlations is noteworthy. This pattern suggests that among IT professionals, knowledge may translate more directly into behavioural change than it does into attitudinal shifts. This could reflect the analytical and problem-solving orientation commonly associated with IT professionals, where factual knowledge may drive practical action more readily than emotional or attitudinal change. This finding is partially consistent with a previous study on pertussis KAP among university students, which also demonstrated significant positive correlations between all KAP domains [9]. However, it contrasts with studies among healthcare workers where high knowledge did not necessarily lead to improved practice [10], possibly because IT professionals, unlike healthcare workers, are encountering this health information freshly and may be more motivated to act on newly acquired knowledge.

The relatively weaker correlation between attitude and practice ($\rho=0.240$) warrants attention. It suggests that positive attitudes towards oral health do not always translate into corresponding practices. This attitude-behaviour gap has been well documented in health psychology literature [11] and may be particularly pronounced in high-pressure work environments where time constraints and competing priorities can prevent individuals from acting on their health beliefs.

4.5 Demographic Associations with KAP Scores

The analysis of KAP scores across demographic variables revealed three statistically significant associations, each providing distinct insights into the factors influencing periodontal health awareness and behaviour in this population.

Gender was the most strongly associated demographic variable, with a highly significant difference in knowledge scores (Kruskal-Wallis $H=42.1$, $p<0.001$). Female respondents demonstrated substantially higher mean knowledge scores (7.7 ± 2.0) compared to males (5.7 ± 2.2). This gender difference in health knowledge is consistent with a large body of literature demonstrating that women tend to possess greater health awareness and engage more actively in health-seeking behaviours [12,13]. In the context of oral health specifically, previous studies have consistently reported that females exhibit higher dental knowledge and more regular dental attendance patterns [14]. However, it is worth noting that despite this knowledge advantage, gender did not significantly influence attitude or practice scores, suggesting that knowledge alone does not account for behavioural differences across genders in this population.

Professional experience was significantly associated with practice scores (Kruskal-Wallis H=18.0, p=0.0012). Notably, professionals with 3–6 years of experience demonstrated the highest practice scores (22.7 ± 3.9), while those in the earliest career stage (0–3 years) had lower scores (20.1 ± 4.1). This pattern does not follow a simple linear progression with experience. The peak in practice scores at the 3–6 year mark may correspond to a career stage where professionals have established stable routines and sufficient income for dental care, yet have not developed the complacency or time pressures that can come with more senior roles. The decline in practice scores among more experienced professionals (10–15 and 15+ years) may also reflect generational differences in oral health education, or the increasing work demands and managerial responsibilities associated with senior positions.

Work mode showed a significant association with attitude scores (Kruskal-Wallis H=11.7, p=0.0029). Remote workers demonstrated the highest mean attitude scores (29.4 ± 2.1), followed by hybrid workers (28.6 ± 5.0), with full-time office workers showing the lowest attitudes (26.8 ± 4.7). This finding is particularly interesting in the post-pandemic context, where remote and hybrid work arrangements have become increasingly prevalent in the IT industry. The higher attitude scores among remote workers may be explained by several factors: greater autonomy over daily routines, reduced commuting time allowing more attention to self-care, and potentially greater exposure to health information through online sources during work-from-home periods. The lower attitudes among full-time office workers may reflect the time constraints and environmental factors of office-based work that limit attention to personal health matters.

Interestingly, age, which has been reported as a significant factor in several KAP studies [9,15], did not significantly influence any of the three KAP domains in this study (p>0.05 for all). Similarly, previous periodontal treatment history did not significantly affect KAP scores, although practice scores showed a trend toward higher values among those with treatment experience (21.8 ± 3.9 vs. 20.7 ± 4.3, p=0.110). The absence of an age effect may be due to the relatively young and homogeneous age distribution of this sample, with 62.0% of respondents falling in the 20–30 years age group.

4.6 Implications for Practice

The findings of this study carry several practical implications. First, the significant knowledge gaps identified — particularly regarding the fundamental concepts of plaque, calculus, and the primary cause of gum disease — highlight the need for targeted oral health education programs for IT professionals. Given that respondents expressed receptivity to workplace awareness programs (A3), employers and dental professionals could collaborate to develop workplace-based oral health initiatives. Such programs could be integrated into existing corporate wellness frameworks, which are increasingly common in the IT sector.

Second, the low utilisation of interdental cleaning aids and infrequent dental visits identified in this study suggest specific behavioural targets for intervention. Dental professionals should emphasise the importance of flossing and regular dental check-ups when treating IT professionals. Corporate dental benefit programs could also be designed to incentivise preventive visits rather than reactive treatment.

Third, the finding that work mode influences attitudes towards oral health is relevant for organisations designing employee wellness programs. Different approaches may be needed for office-based, hybrid, and remote workers, with particular attention to supporting full-time office workers in maintaining positive oral health attitudes and behaviours.

4.7 Limitations

Several limitations should be considered when interpreting the findings of this study. First, this was a cross-sectional study design, which does not permit causal inference between the variables examined. The correlations observed between KAP domains indicate associations but do not establish directionality or causation.

Second, the study used convenience sampling to recruit participants from Bengaluru, which may limit the generalisability of the findings to IT professionals in other cities or regions of India. The demographic profile of the sample, with a predominance of young (62.0% aged 20–30) and male (68.0%) respondents, may not be representative of the broader IT workforce.

Third, the self-administered nature of the questionnaire introduces the possibility of social desirability bias, where respondents may overreport positive attitudes and practices. This is a common limitation in KAP studies that rely on self-reported data [16].

Fourth, the study did not include a clinical examination to validate self-reported practices or assess actual periodontal status. The discrepancy between self-reported practices and actual oral health outcomes has been documented in previous research [17], and future studies would benefit from correlating KAP data with clinical findings.

Fifth, the knowledge questionnaire used a fixed set of 10 items, which, while covering key aspects of periodontal knowledge, may not capture the full breadth of periodontal health literacy. Additionally, the 4-point Likert scale used for attitude assessment did not include a neutral response option, which may have forced respondents towards a directional response.

Despite these limitations, this study contributes to the limited body of literature on periodontal health KAP among IT professionals and provides a foundation for future research and intervention development targeting this growing occupational group.

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